ELEMENTARY CONCEPTS OF ELECTRICITY

Concept of current:-

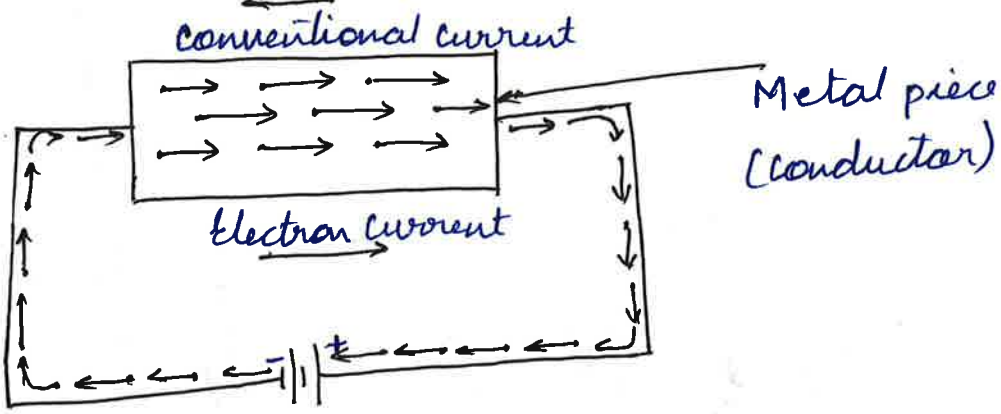
All pure metals are good conductors of electricity, so they contain excess free electrons.

- The flow of these free electrons is nothing but electric current.
- Higher the value of flowing electrons, higher is the value of flow of current.
- Current can also be defined as "rate of flow of charge".

$$\text{Current } (I) = \frac{\text{Charge } (Q)}{\text{Time } (t)}$$

- The unit of current is Ampere (A).

1 Ampere = 6.25×10^{18} electrons.



Electron Current :-

It is the current due to the flow of electrons.

- Electron current flows from -ve of the battery towards +ve end of the battery.

Conventional Current :-

It is the current which is flowing from +ve terminal of battery to -ve terminal of the battery.

- The direction of flow of conventional current is opposite to that of electron current.

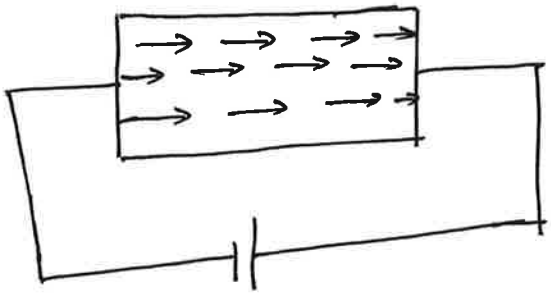
Definition of Ampere :-

A current of one ampere is said to be flowing if a charge of one coulomb passes any given point in a conductor in one second.

- When 6.25×10^{18} electrons pass by a point in a conductor in one second, a current of one ampere is said to be flowing.

$\therefore$ 1A current corresponds to flow of 6.25×10^{18} electrons.

Concept of EMF:-



- It is force required to drive the current from one point to another in particular direction in an electric circuit.
- It is denoted by E.
- Its unit is volts (V).

Electric Potential:-

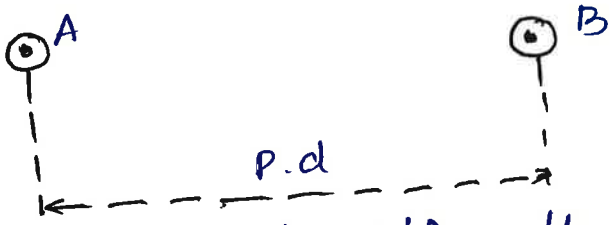
It is the workdone in moving a point charge from one point to closer to the other point against the force of repulsion.

- It is measured in volts.

Potential difference:-

It is the difference between the electric potential at those points.

- It can also be defined as the workdone in moving a point charge from one point to the other against the force of attraction or repulsion.



$V_{AB} = (V_A - V_B)$ volts.

- If potential at A is higher than potential at B, i.e., $V_A > V_B$ then p.d is +ve
- If potential at B is higher than potential at A, i.e, $V_A < V_B$ then p.d is −ve

Concept of Resistance:-

Resistance:-

Resistance of a material is defined as the opposition to the flow of current.

- It is denoted by R & is measured in ohms(Ω).

Mathematical expression:-

$R \propto \frac{l}{a}$, $R = \frac{\rho l}{a}$

$\rho \rightarrow$ Resistivity of a material & it is a constant.

$l \rightarrow$ length of the conductor

$a \rightarrow$ area of cross-section

Factors affecting the value of resistance:-

i) Length of conductor:-

As $R \propto l$, resistance increases with the increase in length.

ii) Cross sectional area:-

As $R \propto 1/a$, resistance decreases with the increase in cross sectional area.

- Resistance of thick conductor is less.

iii) Type of material :-

Resistivity of material depends on the material used.

iv) Temperature :-

As temperature increases, resistance increases.

Conductance (G) :-

It is the reciprocal of resistance i.e., $G = 1/R$.

- Its unit is seimens (S) or mho ($\mho$).

Resistivity (Specific Resistance) :-

It is the property by virtue of which it opposes the flow of current.

$$\rho = \frac{Ra}{l} = \frac{\Omega \times m^2}{m} = \Omega\, m.$$

Name of material	ρ (Ω-m)
Aluminium cast	2.6×10^{-8}
Standard copper	1.72×10^{-8}
Gold	2.36×10^{-8}
Sta Silver	1.58×10^{-8}
Bronze	3.6×10^{-8}
Lead	22×10^{-8}

- The best conductor is that material which has lower value of specific resistance, hence silver is the best conductor.
- The best insulator is the material which has higher value of specific resistance (insulator).

Conductivity (σ):-

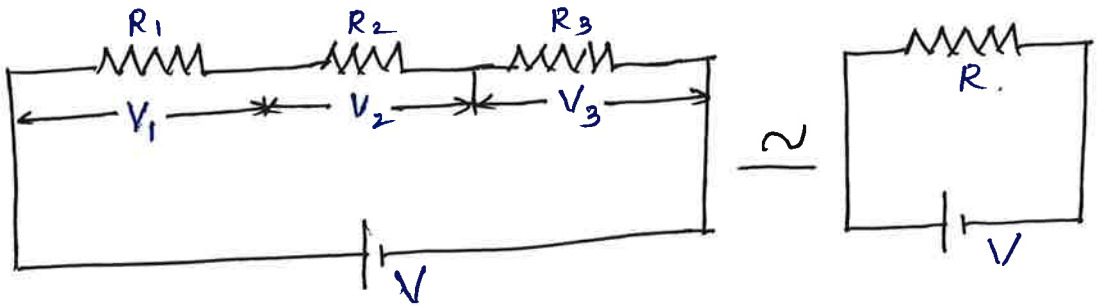
It is the reciprocal of resistivity & is measured in $(\Omega m)^{-1}$ ~~or~~

$$\sigma = 1/\rho .$$

- The conductivity of good conductors such as metal is high & that of insulator such as mica, glass etc is low.

Resistors in series:-

- In this resistors are connected end to end. It is also known as cascade connection.



- All the elements connected in series will carry same current (I).

- By Ohm's law $V = IR$

Here voltage is devided

$$\therefore V = V_1 + V_2 + V_3 + \ldots$$

$$V = I_1R_1 + I_2R_2 + I_3R_3$$

As current is same $I = I_1 = I_2 \ldots$

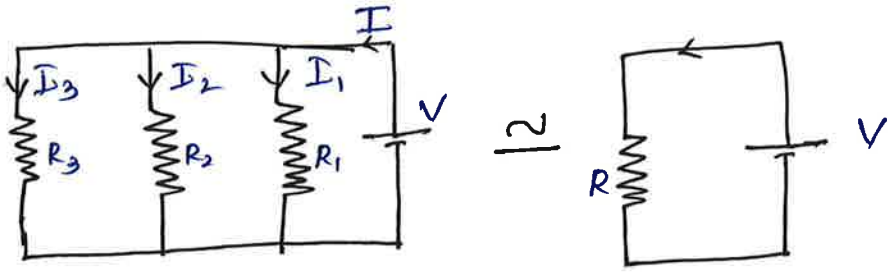
$$\therefore V = I(R_1 + R_2 + R_3 + \ldots)$$

$$\frac{V}{I} = R_1 + R_2 + R_3 \ldots$$

$$R_T = R_1 + R_2 + R_3 \ldots$$

Resistors in parallel:-

- In this the resistors are connected across each other.



- One end of each resistor is connected at a junction.
- Here voltage across each resistor is same, where as current is devided.

$$\therefore I = I_1 + I_2 + I_3 \ldots$$

$$I = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \ldots$$

As voltage is same $V = V_1 = V_2 \ldots$

$$\therefore I = V\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right] \ldots$$

$$\frac{I}{V} = \left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \ldots\right]$$

$$\frac{1}{R} = \left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \ldots\right]$$

Effect of temperature on resistance :-

i) **Pure metals :-**

As the conductors ~~are~~ has more number of free electrons, when such conductor is connected across some voltage, ions are formed inside the conductor & the electrons are moving randomly & opt alligned in a certain direction.

- At low temperature ions are stationary, but as soon as temperature increases, the immovable ions gain energy & start oscillating about their mean position.
- As the temperature increases, the magnitude of oscillation increases.
- This will cause obstruction to flow of free electrons which leads to increase in resistance.

∴ Conductors have +ve temperature co-efficient of resistance.

ii) **Insulators :-**

- Insulators will have less number of free electrons
- When temperature increases vibration of ions increases & simultaneously the electrons gain extra energy from the atom & made available as free electrons.

This will reduce the resistance.

∴ Insulators have −ve temperature co-efficient of resistance.

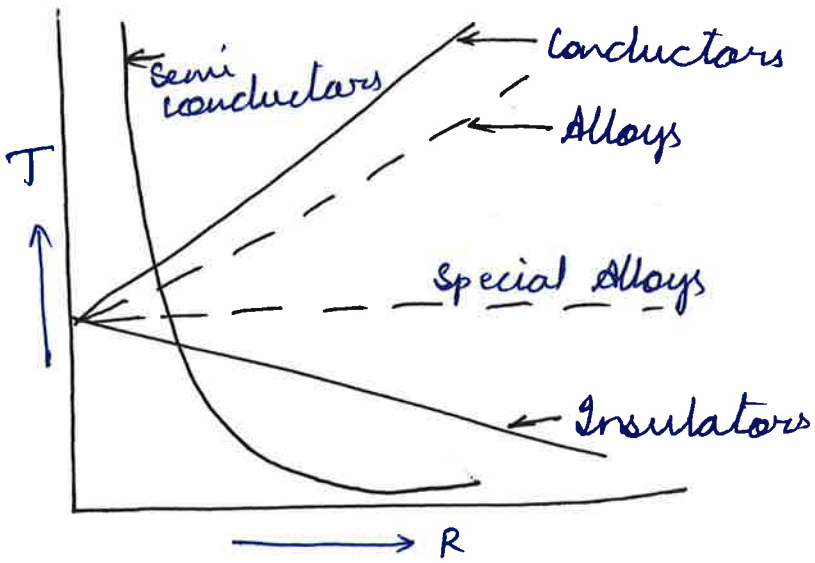
iii) Semi Conductors :-

- At low temperature resistance of semiconductor is high.
- But at high temperature the electrons from valance band will overcome the energy gap & enter into conduction band & become free for conduction.
- So as the temperature increases, resistance decreases.

∴ Semiconductors have -ve temperature co-efficient of resistance.

iv) Alloy :-

- The resistance of alloy increases with increase in temperature, but the rate of increase is not significant.
- Alloy like manganin, eureka show almost no change in resistance with change in temperature
- So such type of alloys are used to make resistance boxes.
- Alloys have +ve temperature co-efficient of resistance



Insulation Resistance :-

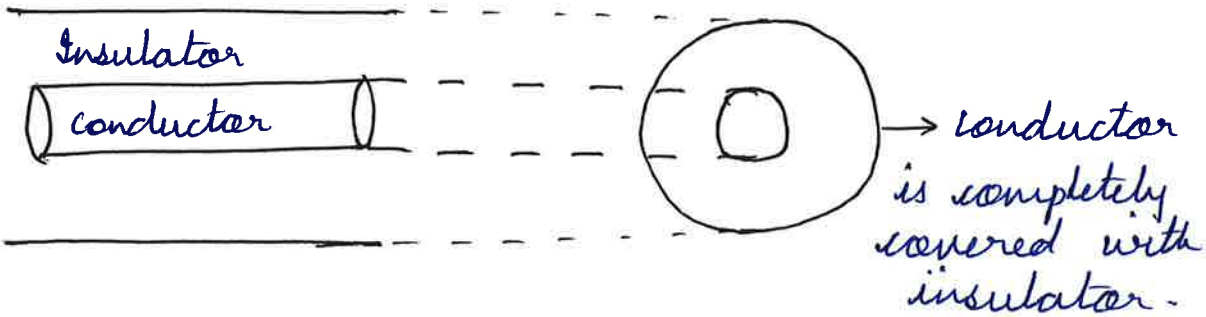
It is the resistance offered by an insulator to the flow of leakage current.

- It is denoted by $R_i = \frac{V}{I_l}$

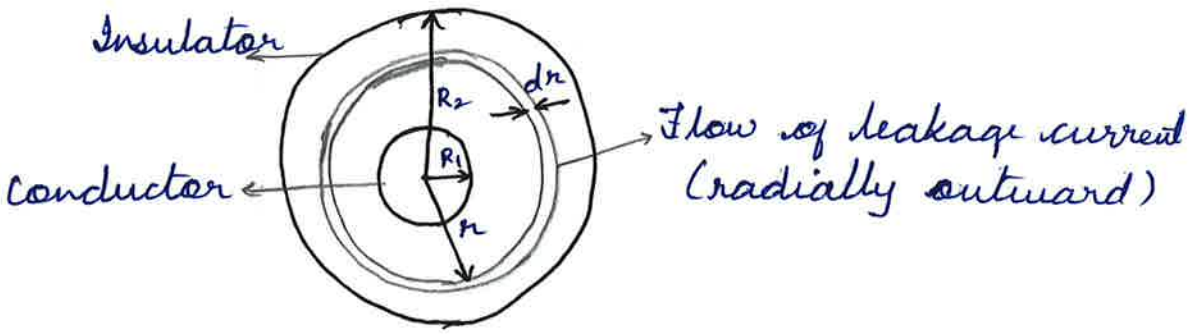
where V = voltage measured between conductor & earth.

I_l = Leakage current

R_i = Insulation resistance.



Expression for insulation resistance of a cable :-



- Due to finite insulation resistance of the insulating material used, some leakage current flows from the conductor in radially outward direction.
- So, to obtain the expression for R_i, we will have to consider a small section of radius 'r' of the cable.
- Let the thickness of this small section be 'dr'.

- As the leakage current flows radially outwards, we have to calculate a small leakage resistance dR_i, we will substitute

Length $= dr$

Cross-sectional area $= 2\pi r l$.

$$dR_i = \rho \times \frac{\text{Length of path followed by leakage current}}{\text{Area}}$$

$$dR_i = \rho \frac{dr}{2\pi r l}$$

where, l = length of the cable.

- The total insulation resistance R_i can be obtained by integrating 'dR_i' over the entire radius of insulating material i.e., from R_1 & R_2

$$R_i = \int_{R_1}^{R_2} dR_i = \int_{R_1}^{R_2} \frac{\rho\, dr}{2\pi r l} = \frac{\rho}{2\pi l} \int_{R_1}^{R_2} \frac{dr}{r}$$

$$R_i = \frac{\rho}{2\pi l} \left[\ln r\right]_{R_1}^{R_2}$$

$$= \frac{\rho}{2\pi l} \left[\ln R_2 - \ln R_1\right]$$

$$\boxed{R_i = \frac{\rho}{2\pi l} \ln\left[\frac{R_2}{R_1}\right] \Omega}$$

where R_1 is radius of conductor

R_2 is radius of cable including conductor & insulator.

* Conclusion! -

- The insulation resistance R_i is inversly proportional to the length 'l' of the cable.
- The insulation resistance R_i is directly proportional to the logarithm of radius of insulating material $[R_2 - R_1]$.
- Insulation resistance is directly proportional to the specific resistance of the insulating material.

* Effect of temperature on insulation resistance: -

- For most of the insulating materials, the resistance decreases with increase in temperature. This happens because, when the temperature of the material increases, some of the valence electron jump into the conduction band & become free for conduction.
- Hence conductivity increases with increase in temperature & the insulation resistance will decrease.
- Mathematically this phenomena can be explained as follows. $\rho_1 = \rho_0 e^{-\alpha_0 T_1}$

where $\rho_1 \rightarrow$ resistivity at temperature T_1
$\rho_0 \rightarrow$ resistivity at temperature $0°c$
$\alpha_0 \rightarrow$ RTC at $0°c$.

* Effect of moisture on insulation resistance! -

As water is good conductor of electricity, if an insulating material absorbs some moisture then it provides a path for the leakage current to flow & insulation resistance will decrease

Numericals on insulation resistance :-

1) A single core cable has 1.5 cm diameter conductor & the thickness of insulation is 2.2 cm. The resistivity of insulating material is $9.2 \times 10^{12}\ \Omega m$. Determine the insulation resistance per km length of the cable. If the working voltage of the conductor is 1100 V, what is the leakage current per km length of the cable?

→ Given :-

Conductor diameter = 1.5 cm

$\therefore$ radius $= R_1 = 0.75 \times 10^{-2}$ m.

$(R_2 - R_1)$ = Thickness of insulation $= 2.2 \times 10^{-2}$ m.

$R_2 = 2.2 \times 10^{-2} + R_1 = 2.2 \times 10^{-2} + 0.75 \times 10^{-2}$.

$R_2 = 2.95 \times 10^{-2}$ m.

we have $R_i = \frac{\rho}{2\pi l} \ln\left[\frac{R_2}{R_1}\right]$

$$= \frac{9.2 \times 10^{12}}{2\pi \times 1000} \ln\left[\frac{2.95 \times \cancel{10^{-2}}}{0.75 \times \cancel{10^{-2}}}\right]$$

$= 2 \times 10^{9}\ \Omega$ per km.

Leakage current for 1 km length

$$I_l = \frac{V}{R_i} = \frac{1100}{2 \times 10^{9}}$$

$I_l = 0.55 \times 10^{-6}$ A.

$I_l = 0.55\ \mu A$.

2) A single core copper cable has conductor diameter of 2 cm with an insulation of thickness 1.8 cm. The resistivity of copper is $1.73 \times 10^{-8}\ \Omega m$ & that of insulation is $8 \times 10^{12}\ \Omega m$. Determine

i) Resistance of conductor.

ii) Insulation resistance if cable length is 100 m.

→ Given:-

Conductor diameter = 2 cm.

$\therefore$ Radius $= R_1 = 1 \times 10^{-2}$ m.

Insulation thickness = 1.8 cm. $= R_2 - R_1$

$$R_2 = 1.8 \times 10^{-2} + R_1$$

$$= 1.8 \times 10^{-2} + 1 \times 10^{-2}$$

$$= 2.8 \times 10^{-2}\ m$$

$$\rho_c = 1.73 \times 10^{-8}\ \Omega m.$$

i) Resistance of conductor

$$R_c = \frac{\rho_c l}{a} = \frac{1.73 \times 10^{-8} \times 100}{\pi (R_1)^2}$$

$$a = \pi \times (1 \times 10^{-2})^2 = 3.14 \times 10^{-4}\ m^2$$

$$\therefore R_c = \frac{1.73 \times 10^{-8} \times 100}{3.14 \times 10^{-4}} = 0.0055\ \Omega.$$

ii) Insulation resistance:-

$$R_i = \frac{\rho_i}{2\pi l} \ln\left[\frac{R_2}{R_1}\right] = \frac{8 \times 10^{12}}{2\pi \times 100} \ln\left[\frac{2.8 \times 10^{-2}}{1 \times 10^{-2}}\right]$$

$$R_i = 1.31 \times 10^{10}\ \Omega.$$

3) A single core cable has its diameter of conductor as 1.5 cm & outer diameter as 3.9 cm. The resistivity of conductor & insulation are $1.73 \times 10^{-8}\ \Omega m$ & $8 \times 10^{12}\ \Omega m$ respectively. Find for a cable length of 100m its insulation resistance & conductor resistance.

→ Given :-

Conductor diameter $= 1.5 \times 10^{-2}$ m

$R_1 = 0.75 \times 10^{-2}$ m

Cable diameter $= 3.9 \times 10^{-2}$ m

$R_2 = 1.95 \times 10^{-2}$ m

$\rho_c = 1.73 \times 10^{-8}\ \Omega m$

$\rho_i = 8 \times 10^{12}\ \Omega m$

$l = 100$ m.

i) Conductor resistance (R_c) :-

$$R_c = \frac{\rho_c l}{a} = 1.73 \times 10^{-8} \times \frac{100}{\pi (R_1)^2}$$

$$a = \pi (R_1)^2 = \pi \times (0.75 \times 10^{-2})^2$$

$$a = 3.14 \times (0.75 \times 10^{-2})^2$$

$$R_c = 1.73 \times 10^{-8} \frac{\times 100}{3.14 \times (0.75 \times 10^{-2})^2}$$

$$R_c = 9.789 \times 10^{-3}\ \Omega.$$

ii) Insulation resistance (R_i)

$$R_i = \frac{\rho_i}{2\pi l} \ln\left[\frac{R_2}{R_1}\right] = \frac{8 \times 10^{12}}{2\pi \times 100} \ln\left[\frac{1.95 \times \cancel{10^{-2}}}{0.75 \times \cancel{10^{-2}}}\right]$$

$$R_i = 1.2165 \times 10^{10}\ \Omega.$$

4) The insulation resistance per km of a single core cable having a conductor diameter of 1.2 cm & insulation thickness of 1.5 cm is 550 MΩ. Find percent increase in insulation resistance if insulation thickness is increased by 50%.

→ $R_i = 550\ M\Omega/km = 550 \times 10^6\ \Omega/km$

$d_1 = 1.2\ cm \quad \therefore \quad R_1 = 0.6 \times 10^{-2}\ m.$

$R_2 - R_1 = 1.5 \times 10^{-2}.$

$R_2 = 1.5 \times 10^{-2} + R_1 = 1.5 \times 10^{-6} + 0.6 \times 10^{-6}.$

$R_2 = 2.1 \times 10^{-2}\ m.$

$$R_i = \frac{\rho}{2\pi L} \ln\left[\frac{R_2}{R_1}\right]$$

$$550 \times 10^6 = K \ln\left[\frac{2.1 \times \cancel{10^{-2}}}{0.6 \times \cancel{10^{-2}}}\right]$$

$$K = 439 \times 10^6.$$

New thickness of insulation $= 1.5 + \frac{1.5}{2} = 1.5 + 0.75$

$= 2.25\ cm$

$R_2 = (0.6 + 2.25) \times 10^{-2} = 2.85 \times 10^{-2}\ m.$

$$R_i = K \ln\left[\frac{R_2}{R_1}\right] = 439 \times 10^6 \ln\left[\frac{2.85 \times \cancel{10^{-2}}}{0.6 \times \cancel{10^{-2}}}\right]$$

$R_i = 684.025\ M\Omega = 684.025 \times 10^6\ \Omega$

$$\% \text{ increase in } R_i = \frac{(684.025 - 550) \times \cancel{10^6}}{550 \times \cancel{10^6}} \times 100$$

$= 24.37\%$

WORK, POWER, AND ENERGY

1) Thermal System:-

a) Joule's law:-

It states that the amount of electrical energy spent is equal to the amount of heat energy produced. The heat produced is due to the current flowing through a material.

Electrical energy expended = Heat produced.

or

The amount of heat produced is directly proportional to the square of the current, is directly proportional to the resistance of the conducting material & is directly proportional to the time in which electric current is passed through the conducting material.

$H \propto I^2 R t$.

b) Calorie:-

It is the amount of heat required to raise the temperature of one gram of water through 1°c.

1 calorie = 4.186 joules

1 Joule = 1/4.186 calories

= 0.2389 calories

1kWh = 3.6×10^6 J

= 860 kCal.

c) Heat capacity:-

The amount of heat required to produce a given temperature rise in a given mass of material.

- It can also be defined as the ratio of amount of heat supplied to the corresponding rise in temperature.

$$\text{Heat capacity} = \frac{Q}{T_2 - T_1} \text{ Joules / Kelvin}$$

Where $Q \rightarrow$ Heat supplied in Joules.
$T_2 - T_1 \rightarrow$ change in temperature.

d) Specific heat :-

It is defined as heat capacity of the body per unit mass.

- If mass of the body is 'm' kg, specific heat is given by $s = \frac{Q}{m(T_2 - T_1)} = J / kg\,K$.

e) Sensible heat :-

It is the amount of heat gained or lost due to change in temperature.

- If a body is heated from temperature T_1 to T_2 then the heat is supplied or gained & if a body is cooled from T_2 to T_1, then the heat is lost by the body.
- If 'm' is the mass of the body.
  s is specific heat.
  T_1 & T_2 are temperature, then sensible heat is given by.

$$\text{Sensible Heat} = \text{mass} \times \text{specific heat} \times \text{change in temp.}$$
$$= m s (T_2 - T_1)$$
$$= m s \Delta T$$

f) Latent heat :-

It is the amount of heat energy required to change the form of the substance.

- When the conversion takes place, the temperature remains constant at its melting point, hence heat is not sensed by the body, in terms of temperature rise. Hence it is called latent heat.

i) Latent heat of liquification :-

It is the heat required to convert a body from ~~sto~~ solid state to liquid state.

ii) Latent heat of fusion :-

It is the heat required to convert liquid state to gaseous state.

$$\text{Latent heat} = m \times L$$

where 'm' is mass in kg.

L is latent heat in J/kg.

$\therefore$ Total heat = Sensible heat + Latent heat

$$= MS\Delta T + m \times L$$

g) Calorific value :-

It is the amount of heat energy produced in joules or kilo joules by burning the unit mass of fuel (such as coal, petrol, diesel etc).

- It is measured in J/kg.

$\therefore$ Heat energy produced = Mass × Calorific value

$$= \cancel{kg} \times \frac{J}{\cancel{kg}} = \text{Joules}$$

h) Water equivalent of a container :-

While heating water, its container is also heated, this is waste of heat. To take wastage into account, the container body is assumed to be equivalent to some quantity of water which is called as water equivalent of container.

i) Radiation loss :-

The heat energy escaping to the surroundings due to radiation is called as radiation loss.

- It is measured in Joules.

Numericals :-

1) An immersion heater supplied at 250V dc is placed in a bucket containing 5 litres of water. The temperature of water goes up by 50°C in 20 minutes. The water equivalent to the bucket is 500g. Calculate the heater resistance. Specific heat of water is 4180 J/kg K & heat lost due to radiation is negligible.

→ Given :-

$V = 250V$, mass of water $= 5l = 5kg$, $\Delta T = 50°C$

$t = 20 min = 20 \times 60 = 1200$ seconds

mass of bucket $= 500g = 0.5$ kg (water equivalent of a container)

$S = 4180 J/kg K$.

$\therefore$ Total mass $= 5kg + 0.5kg = 5.5kg$.

Op energy $E_o = mS\Delta T$

$= 5.5 \times 4180 \times 50$

$= 1149500$ Joules.

$E_i = E_o$ (radiation losses are negligible)

$\therefore E_i = 1149500 J$

$$P_{in} = \frac{E_{in}}{time} = \frac{1149500}{20 \times 60} = 957.91\ W.$$

$$P_{in} = \frac{V^2}{R}, \quad R = \frac{V^2}{P_{in}}$$

$$R = \frac{(250)^2}{957.91} = 65.246\ \Omega$$

$$R = 65.246\ \Omega.$$

2) An electric heater is required to heat 30 litres of water from 12°c to 100°c. Calculate the electric energy consumed in kWh if the efficiency of heater is 85% & specific heat of water is 4190 J/kg K.

→ $m = 30\,l = 30\,kg$, $T_2 = 100°C$, $T_1 = 12°C$, $\eta = 85\%$.
$s = 4190\ J/kg\ K$.

$$E_o = ms\Delta T = 30 \times 4190 \times (100 - 12)$$
$$= 11.0616 \times 10^6\ J$$

$$E_{in} = \frac{E_o}{\eta} = \frac{11.0616 \times 10^6}{0.85} \qquad \eta = \frac{E_o}{E_{in}}$$

$$E_{in} = 13.0136 \times 10^6\ J$$

$$1\ kWh = 36 \times 10^5\ J$$

$$\therefore E_{in} \text{ in kWh} = \frac{13.0136 \times 10^6}{36 \times 10^5}$$

$$= 3.615\ kWh.$$

3) A bucket contains 20 litres of water at 20°C. A 2.5 kW immersion heater is used to raise temperature of water to 95°C. If the overall efficiency of process is 90%, calculate the time required for the process. Find the cost of energy for 365 days if energy rate is Rs 7 per unit. Assume specific heat capacity of water to be 4200 J/kg K.

→ Given :-

$m = 20l = 20kg$, $s = 4200 J/kg K$, $P_{in} = 2.5 kW = 2.5 \times 10^3 W$

$T_1 = 20°C$, $T_2 = 95°C$, $\eta = 90\% = 0.9$.

$$E_o = ms\Delta T = 20 \times 4200(95-20)$$

$$= 63 \times 10^5 J$$

$$E_{in} = \frac{E_o}{\eta} = \frac{63 \times 10^5}{0.9}$$

$$= 7 \times 10^6 J$$

$$\text{Power} = \frac{\text{Energy}}{\text{Time}}, \quad \text{Time} = \frac{\text{Energy}}{\text{Power}}.$$

$$\text{Time} = \frac{7 \times 10^6}{2.5 \times 10^3}.$$

$$= 2800 \text{ seconds}$$

$$= 46.6 \text{ minutes}.$$

E_{in} in kWh [∵ 1 unit = 1 kWh]

$$1 kWh = 36 \times 10^5 J$$

$$E_{in} = \frac{7 \times 10^6}{36 \times 10^5} = 1.94 kWh = 1.94 \text{ units}$$

Bill per year = No. of units × cost/unit × no. of days

$$= 1.94 \times 7 \times 365$$

$$= 4956.7 \text{ Rs}.$$

4) An electric kettle is required to raise the temperature of 2 litres of water from 20°C to 100°C in 15 minutes. Calculate the resistance of heating element if the kettle is to be used on 200V supply. Assume efficiency of kettle to be 80%. Assume specific heat of water as 4180 J/kg K

→ Given:-

$m = 2l = 2kg$, $s = 4180\ J/kg\ K$, $T_1 = 20°C$, $T_2 = 100°C$

$t = 15\ min = 15 \times 60 = 900$ seconds

$\eta = 80\%$, $V = 200V$, $R = ?$

$$E_o = ms\Delta T$$
$$= 2 \times 4180 \times (100 - 20).$$
$$= 668800 \text{ Joules.}$$

$$E_{in} = \frac{E_o}{\eta} = \frac{668800}{0.8}$$
$$= 836 \times 10^3\ J$$

$$P_{in} = \frac{E_{in}}{time} = \frac{836 \times 10^3}{900}$$
$$= 928.88\ W.$$

$$P_{in} = V \times I = \frac{V^2}{R}.$$

$$R = \frac{V^2}{P} = \frac{(200)^2}{928.88} = 43.06\ \Omega$$

$$R = 43.06\ \Omega$$

5) How long it takes to raise the ~~the~~ temperature of 880g of water from 16°C to boiling point? The heater takes 2A at ~~20~~ 220V supply & has an efficiency of 90%. Assume the specific heat of water to be 4190 J/kg K. & 1 litre of water to have a mass of 1 kg.

→ $m = 880g = 0.88\ kg$, $T_1 = 16°C$, $T_2 = 100°C$, $I = 2A$
$V = 220V$, $\eta = 0.9$, $s = 4190\ J/kg\ K$.

$$E_o = ms\Delta T = 0.88 \times 4190 (100 - 16)$$
$$= 309742.8 \text{ Joules.}$$

$$E_{in} = \frac{E_o}{\eta} = \frac{309742.8}{0.9}$$
$$= 344138.67 \text{ Joules.}$$

$$\text{Power} = \frac{E_{in}}{\text{Time}} \qquad \text{Time} = \frac{E_{in}}{\text{Power}}$$

$$t = \frac{344138.67}{V \times I} = \frac{344138.67}{200 \times 2}$$

$$t = 782.13 \text{ seconds}$$
$$= 13.03 \text{ minutes.}$$

6) An electric furnace is used to melt 1000 kg of tin per hour. Input to the furnace is 50 kW. Melting point of tin is 235°C. Latent heat of liquification for tin is 13.3 k cal/kg. Specific heat is 231 J/kg K & the initial temperature is 15°C. Calculate the efficiency of the furnace.

→ $P_{in} = 50\ kW = 50 \times 10^3\ W$, $m = 1000\ kg$, $s = 231\ J/kg\ K$,
$T_1 = 15°C$, $T_2 = 235°C$, $L = 13.3\ kCal = 13.3 \times 10^3 \times 4.186$
$\therefore L = 55673.8\ J/kg$

E_o = Sensible heat + Latent heat

$= ms\Delta T + m \times L$

$= 10^3 \times 231 (235-15) + 100 \times 55673.8$

$= 106.49 \times 10^6$ J/hr.

~~$E_{in} = \frac{E_o}{\eta} =$~~

$P_o = \frac{E_o}{time} = \frac{106.49 \times 10^6}{3600}$ J/s or Watt.

$P_o = 29580.55$ W.

$\eta = \frac{P_o}{P_{in}} \times 100 = \frac{29580.55}{50 \times 10^3} \times 100$

$\eta = 59.16\%$

7) An electric furnace is used to melt aluminium. Initial temperature of solid aluminium is 32°C & its melting point is 680°C. Specific heat capacity of aluminium is 0.95 kJ/kg K & heat required to melt 1kg of Al at its melting point is 450 kJ. If the input power drawn by the furnace is 20 kW & its overall efficiency is 60%. Find the mass of Al melted per hour

→ $T_1 = 32°C$, $T_2 = 680°C$, $S = 0.95 \times 10^3$ J/kg K.

Heat required to melt 1kg of Al at 680°C = $L = 450$ kJ

$= 450 \times 10^3$ J

$P_{in} = 20$ kW $= 20 \times 10^3$ W, $\eta = 0.6$

$E_o = ms\Delta T + mx\ell = m[s\Delta T + \ell]$

$= m[0.95\times10^3(680-32) + \cancel{0.9}\,450\times10^3]$

$= 1065600\times m$ Joules

$E_{in} = \frac{E_o}{\eta} = \frac{1065600\,m}{0.6}$

$= 177600\,m$ Joules.

$E_{in} =$ Power $\times$ time

$= 20\times10^3\times3600$

$= 72\times10^6$ Joules

$177600\,m = 72\times10^6$

$m = \frac{72\times10^6}{1776000}$

$m = 40.54$ kg.

8) A boiler steam turbine generator system consumes 1000 tons of coal & generates 2000 MWh energy in a day. The calorific value of the coal is 30 MJ/kg Find the overall efficiency of the system.

→ $E_o = 2000$ MWh $= 2000\times10^6$ Wh.

Calorific value $= 30$ MJ/kg $= 30\times10^6$ J/kg

$m = 1000$ ton $= 1000\times10^3$ kg $= 10^6$ kg.

$E_{in} =$ mass $\times$ calorific value

$= 10^6\times30\times10^6$

$= 30\times10^{12}$ J

~~$P_{in} = 833$~~ $P_{in} = \frac{E_{in}}{time} = \frac{30\times10^{12}}{3600}$

$P_{in} = 8333.33$ MWh

$\therefore \eta = \frac{O/P}{I/P} = \frac{2000\times\cancel{10^6}}{8333.33\times\cancel{10^6}} = 24\%$

Mechanical System:-

a) **Mass:-**

The matter possessed by a body (m).

- Its unit is kg.

b) **Velocity (v):-**

Distance travelled in unit time or rate of change of displacement is called velocity

- Its unit is metre/second (ms^{-1}).

c) **Acceleration:-**

Rate of change of velocity is called acceleration

$$a = \frac{\text{Velocity}}{\text{time}} = \frac{ms^{-1}}{s}$$

$$= ms^{-2}.$$

d) **Force (F):-**

Force is an external agent which when acting on a body either changes or tends to change the position of the body.

$$\text{Force} = \text{mass} \times \text{acceleration}$$

$$= kg\ ms^{-2}$$

$$= \text{Newton (N)}$$

e) **Weight:-**

It is the product of mass & gravitational acceleration.

$$W = m \times g \qquad g = 9.81\ ms^{-2}$$

f) **Torque:-**

Torque is the product of force & ~~to~~ distance between the line of action of force & the axis of rotation.

$$T = F \times r, \quad F \text{ is force}$$
$$r \text{ is } \cancel{to} \text{ distance or radial distance}$$

- Its unit is N-m.

g) **Work:-**

When a force F acts on a particle, then the particle moves through a distance d.

- In this situation work is said to be done & it is defined as the product of force & displacement.

Workdone (W) = Force (F) × Displacement (d)

- Its unit is N-m or joule.

h) **Power:-**

It is defined as rate of workdone w.r.t time or rate of doing work.

$$\text{Power (P)} = \frac{\text{Workdone}}{\text{time}}$$

- It is measured in Joules/second or Nm/s or Watt.

i) Energy (E) :-

Capacity to do work is called energy & is measured in joules.

Types of energy :-

i) Kinetic Energy :-

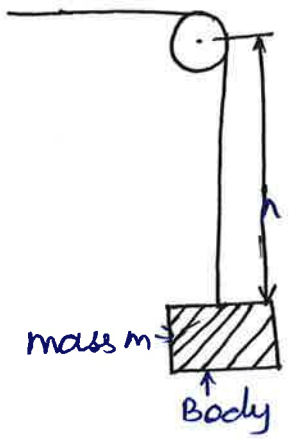
It is the energy possessed by a body due to its motion.

- If a body of mass 'm' is moving with a velocity 'v', then KE is given by

$$KE = \frac{1}{2} m v^2 \text{ Joules.}$$

Potential Energy :-

It is the energy possessed by a body due to its position



When a body of mass 'm' is lifted vertically through a height 'h', the PE is given by,

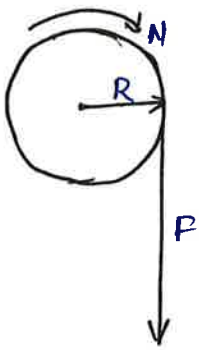
$$PE = mgh = Wh \text{ Joules}$$

Where W is weight of the body.
g is gravitational force.

The relation between power & energy is given by

Energy = Power × Time = Workdone.

Relation between Power & Torque :-



Let us consider a pulley of radius R.
Let F be the force applied to it.
Then, torque $T = F \times R$ Nm.
Let the pulley rotate N revolutions per minute.

∴ Workdone in one revolution is given by.

W = Force × distance travelled in one revolution

- But distance travelled in one revolution is
d = circumference of pulley $= 2\pi R$.

$\therefore W = F \times 2\pi R$ Joules.

- Time corresponding to one revolution $= \frac{60}{N}$ seconds.

$$\therefore \text{Power} = \frac{\text{Workdone}}{\text{Time}} = \frac{F \times 2\pi R}{60/N}$$

$$\text{Power} = F \times R \times \frac{2\pi N}{60}$$

$P = T \times \omega$ (From eq ① $T = F \times R$)

$\omega = \frac{2\pi N}{60}$ is angular velocity in rad/s.

$\therefore P = T \times \omega$

Electrical System:-

a) **Electrical work done:-**

Workdone to transfer a charge from one point to the other.

- The unit of electric workdone is Joules.

* **One Joule:-**

The amount of workdone to move a unit charge (1 coulomb) through the potential difference of 1 volt.

- Electrical workdone can also be defined as workdone to move a unit charge (Q) through a potential difference of V volts.

$$\therefore \text{Workdone} = Q \times V$$

$$= V \times I \times t \text{ Joules} \qquad (Q = It)$$

b) **Power (P):-**

The electric power is due to voltage & current hence it is defined as product of voltage & current

- It can also be defined as rate of doing the electrical work.

$$P = V \times I$$

$$P = \frac{\text{electrical work done}}{\text{time}}$$

$$P = \frac{V \times I \times \cancel{t}}{\cancel{t}}$$

$$\therefore P = V \times I$$

- It is denoted by P & is measured in watts (W).

$$P = V \times I$$

But by ohm's law $V = IR$

$$\therefore P = IR \times I$$

$$= I^2 R \text{ Watts (in terms of current).}$$

Put $I = V/R$.

$$P = V \times \frac{V}{R} = \frac{V^2}{R} \text{ Watts (in terms of voltage).}$$

c) Energy:-

When total electrical work is done in any electrical circuit, energy is said to be spent.

Energy (E) = Total electric workdone

$$E = V \times I \times t$$

$$\therefore E = P \times t$$

- Hence energy can also be defined as product of power & time.
- The SI unit of energy is watt-second or joule.
- The electricity meters supplied by MSEB are actually energy meters. They measure energy consumed by the customer in watt hour or kilo watt hour.

1 kWh = 1 unit.

1) Find the energy & current to a crane driving motor when raising a load of 1 tonne through a height of 16 m in 10 seconds. Motor efficiency & gear efficiency are 85% & 60% respectively & supply voltage is 400V.

→ Given:-

$h = 16\ m,\ t = 10\ s,\ m = 1\ tonne = 1000\ kg$

$\eta_m = \frac{85}{100} = 0.85,\quad \eta_g = \frac{60}{100} = 0.6,\quad V = 400V.$

$E_o = mgh = 10^3 \times 9.81 \times 16$

$= 156960\ N\text{-}m.$

$E_i = \frac{E_o}{\eta_m \eta_g} = \frac{156960}{0.85 \times 0.6}$

$E_{in} = 307764.7\ N\text{-}m.$

$E_{in} = P \times time = V \times I \times t$

$I = \frac{E_{in}}{V \times time} = \frac{307764.7}{400 \times 10}$

$I = 76.944\ A.$

2) A feed pump delivers 15 kilo litres of hot water per hour to a boiler. Efficiency of the pump is 90% & the pump is driven by a 480 V d.c motor having an efficiency of 0.85. Assume hot water to weigh 946 kg/m³ & effective pressure at which the water is delivered is equivalent to a head of 420 m. Determine motor current.

→ Given :-

$1 m^3$ of water = 946 kg.

$\therefore 1 m^3 = 1 kl$.

hence $m = 15 \times 946 = 14190$ kg.

$\eta_p = 0.9$, $\eta_m = 0.85$, $V = 480V$, $h = 420 m$

$t = 1 hr = 60 \times 60 = 3600$ seconds.

$E_0 = mgh = 14190 \times 9.81 \times 420$

$= 58465638 J$.

$$E_{in} = \frac{E_0}{\eta_p \eta_m} = \frac{58465638}{0.85 \times 0.9}$$

$= 76425670.6 J$

$E_{in} = P \times t = V \times I \times t$

$$\therefore I = \frac{E_{in}}{V \times t} = \frac{76425670.6}{480 \times 1 \times 60 \times 60}$$

$I = 44.22 A$.

3) An electric pump lifts $60 m^3$ of water per hour to a height of 25 m. The pump efficiency is 82% & the motor efficiency is 77%. The pump is used for 3 hours daily. Find the energy consumed per week if the mass of $1 m^3$ of water is 1000 kg.

→ Given :-

$1 m^3 = 1000 kg$

$\therefore m = 60 \times 1000 = 60 \times 10^3 kg.$

$h = 25 m, \eta_p = 0.82, \eta_m = 0.77.$

$E_o = mgh = 60 \times 10^3 \times 9.81 \times 25$

$= 14715000 J$

$E_{in} = \frac{E_o}{\eta_m \eta_p} = \frac{14715000}{0.82 \times 0.77}.$

$= 23305353.18 J.$

Energy consumed in kWh $= \frac{23305353.18}{36 \times 10^5}$

$= 6.47$ kWh.

$= 6.47$ unit.

Pump is used 3 hours daily.

$\therefore$ Energy consumed per week $= 6.47 \times 3 \times 7.$

$= 135.94$ units.

4) A three blade wind mill is used to lift underground water & store it at ground level using a pump. For average wind speeds, the value of torque developed is 20 Nm & the speed of this wind mill is 150 rpm. Actual head of water is 9m & pipe friction is 1m head loss. The wind mill mechanical efficiency & water pump efficiency are 40% & 75% respectively. Calculate the run of this wind mill to store water quantity of 20 kl at ground level.

→ Given! –

$T = 20\,Nm$, $N = 150\,r.p.m$, $h = 9+1 = 10\,m$.

$\eta_m = 40\% = 0.4$, $\eta_p = 75\% = 0.75$, $t = ?$

$m = 20\,kl = 20 \times 10^3\,l$.

$E_o = mgh = 20 \times 10^3 \times 9.81 \times 10$

$= 1962000\,J$

$$E_{in} = \frac{E_o}{\eta_m \eta_p} = \frac{1962000}{0.4 \times 0.75}$$

$E_{in} = 6540{,}000\,J$

$E_{in} = P_{in} \times$ time

$P = T \times \omega$, $\omega = \frac{2\pi N}{60}$

$$= 20 \times \frac{2\pi \times 150}{60}$$

$= 314.1592654\,W$.

~~tm~~ time $= \frac{E_{in}}{P_{in}}$

$$= \frac{6540{,}000}{314.1592654}$$

$= 20817.46655$ seconds

$= 346.9577$ minutes

$= 5.782$ hours

Electrical Power System :-

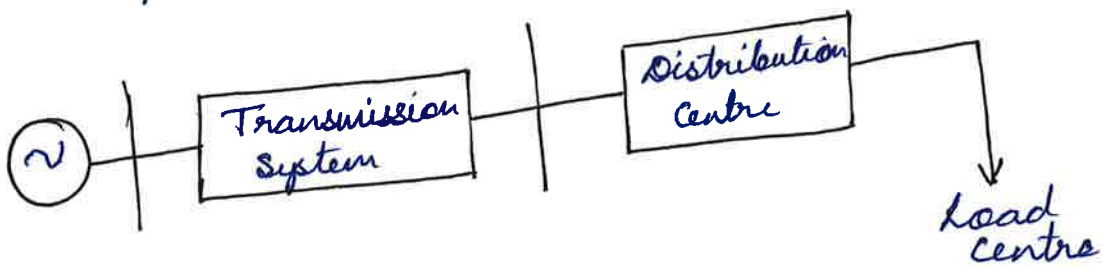
In general, the modern power system consists of following principle elements.

i) Generation

ii) Transmission

iii) Distribution

Block Diagram of Power System :-



- The main objective of an electrical power system is to obtain electrical energy & make it reachable safely to the load point where it is being used in usable forms.
- This is done in following stages.

i) Generation or Generating station :-

The place where electric power produced by parallel connected three phase alternators / generators is called generating station (i.e. power plant).

- The ordinary power plant capacity & generating voltage may be 11 kV, 11.5 kV, 12 kV or 13 kV.

But economically, it is good to step up the produced voltage to 132 kV, 220 kV or 500 kV or more by step up transformer.

- Generation is the part of power system where we convert some form of energy into electrical energy. This is the source of energy in power system. (ex. Thermal, Hydel, Nuclear, Diesel, solar, Tidal wind power plant etc).

- It keeps running all the time. It generates power at different voltage & power levels depending upon the type of station & the generators used.

ii) Transmission :-

a) Primary or High voltage transmission :-

- The electric supply is transmitted to load ~~centre~~ centre by three phase three wire overhead transmission system.
- The transmission lines carry the power at the high voltage from generating station to recieving station which is usually located at outskirts of the city. This forms a primary transmission system.
- The distance between generating station & recieving station is normally very large.

b) Secondary or Low voltage transmission :-

- Area far from the city (outskirts) which are connected with recieving stations by lines is called secondary transmission.

- At recieving station, the level of voltage reduced by step down transformers upto 132 kV, 66 kV or 33 kV, & electric power is transferred by three phase three wire overhead system to different sub stations.

iii) Distribution :-

a) Primary or High voltage distribution :-

- At a sub station, the level of secondary transmission voltage (132 kV, 66 kV or 33 kV) reduced to 11 kV by step down transformers.
- Generally, electric supply is provided to those heavy load consumer (industries) where the demands is 11 kV, from ~~which~~ lines which carries 11 kV & they make a separate substation to control & utilize the heavy power in industries & factories.

b) Secondary or Low voltage distribution :-

- Electric power transferred from primary distribution line to distribution sub station is known as secondary distribution.
- This sub station is located near by domestic & ~~consumer~~ consumer areas where the level of voltage is reduced to 440 V by step down ~~of~~ transformer.